MAHATMA GANDHI UNIVERSITY, KOTTAYAM MGU-UGP (HONOURS) Regular EXAMINATION October 2025

Third SEMESTER Discipline Specific Core Course (DSC) - MG3DSCSAC200 – SPEED, AGILITY, QUICKNESS AND PLYOMETRICS (2024 ADMISSION ONWARDS)

QP Code -25900657

PART A

1. a) Enhances motor skills
2. b) Speed, coordination, Agility
3. a) Speed maintained without oxygen supply
4. a) Shuttle Run
5. a) Middle distance running
6. a) Multidirectional movement
7. d) improves explosive movement more effectively
8. b) Restore functional movement
9. a) Improve horizontal explosive power
10. d) Jumping and sprinting

PART B

11. Speed is the ability to move the body or a body part from one point to another as fast as possible. It mainly focuses on how quickly a person can cover a distance or perform a movement.

Quickness, on the other hand, is the ability to react and move rapidly in response to a stimulus. It involves reaction time and fast movement changes rather than just running speed.

12. SAQ (Speed, Agility, and Quickness) training is more effective than steady jogging for weight loss because it involves high-intensity movements that engage multiple muscle groups and elevate the heart rate rapidly. This leads to greater calorie burn in a shorter time compared to steady jogging. SAQ training also improves metabolism.

13. Agility is the ability to move and change direction quickly and efficiently while maintaining balance, coordination, and control of the body.

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14. Rope drills (such as ladder or jump rope exercises) primarily improve the following components of fitness:
    - Agility – by training quick foot movements and rapid direction changes.
    - Coordination – through precise timing between the eyes, hands, and feet.
    - Balance – by maintaining body control during fast, repetitive movements.
    - Speed – by enhancing reaction time and movement efficiency.

15. Two key benefits of performing drop jumps are:

- They develop **explosive leg power** by training the muscles to generate force quickly during landing and take-off.
- They improve **reactive strength and landing mechanics**, enhancing control, stability, and injury prevention during dynamic movements.

16. **Acceleration speed** is the ability to increase movement velocity rapidly from a stationary or slow position. It focuses on how quickly an athlete can reach their top speed — usually important in the initial phase of a sprint or quick starts in sports.

    Maximum speed, on the other hand, is the highest velocity an athlete can maintain once full acceleration has been achieved. It is typically sustained for a short duration before fatigue sets in.

17. The benefits of bunny jumps include:

- Improves leg power and strength – strengthens the muscles of the legs, especially the quadriceps, hamstrings, calves, and glutes.
- Enhances coordination and balance – trains the body to move rhythmically and maintain stability during repetitive jumps.
- Increases muscular endurance – helps the legs sustain repeated explosive movements over time.
- Develops explosive ability – improves the stretch-shortening cycle, making movements quicker and more powerful.

18. Plyometric training enhances change of direction ability by improving the muscles' ability to produce and absorb force quickly. When performing plyometric exercises, the muscles undergo rapid eccentric (stretching) and concentric (shortening) actions, which train the stretch-shortening cycle**.** This helps athletes decelerate, stabilize, and re-accelerate efficiently when changing direction.
19. Exercise selection plays a crucial role in determining the effectiveness of plyometric training because it ensures the exercises match the athlete's goals, sport demands, and fitness level**.** Choosing the right plyometric exercises targets specific muscle groups and movement patterns that are relevant to the sport or activity. For example, vertical jumps benefit basketball players, while bounding drills are better for sprinters who need horizontal power.

20. Progression ensures that training is safe, structured, and effective, leading to continuous improvement without risking injury.

21. **Resistance Bands:**
    - Provide variable resistance — the tension increases as the band stretches.
    - Help improve acceleration, control, and explosive speed during jumps or throws.
    - Are useful for developing coordination and technique because resistance changes through the movement.
    - Reduce impact stress during landing, making them safer for beginners.

    Weighted Vests:
    - Add constant external load to the body, increasing overall resistance throughout the movement.
    - Help build strength, power, and endurance by overloading the muscles.
    - Are effective for athletes with a solid strength base who want to increase jump height or sprint power.

22. Proper surface selection helps absorb shock, prevent injuries, and ensure safe, effective performance during plyometric training. Using soft, shock-absorbing

surfaces—like grass, rubber mats, wooden floors, or synthetic tracks—helps reduce impact, improve landing safety, and protect the lower limbs.

PART C

23. Speed, Agility, and Quickness (SAQ) training plays a vital role in rehabilitation by helping individuals regain functional movement efficiency after an injury. It focuses on re-educating the body to perform controlled, explosive, and coordinated movements essential for returning to normal or sport-specific activities.

    During rehabilitation, SAQ drills help to restore neuromuscular coordination, balance, and reaction time, which are often reduced due to injury or immobilization. These exercises also enhance joint stability and muscle responsiveness, minimizing the risk of re-injury.

    Moreover, SAQ training improves confidence and body awareness, allowing a smoother transition from basic therapy to sport readiness. It bridges the gap between rehabilitation and performance, ensuring that the athlete can safely return to competitive or daily functional movement.

24. Using MAS profiling allows coaches to customize training loads, ensuring that each football player develops the speed, power, and endurance needed for optimal match performance while avoiding overtraining. Maximum Anaerobic Speed (MAS) is the lowest running speed at which an athlete reaches their maximal oxygen uptake (VO_2 max). It is a valuable measure used to design individualized conditioning programs that improve both aerobic and anaerobic capacity in football players.

    By determining each player's MAS through tests such as the Yo-Yo Intermittent Recovery Test or Time Trials, coaches can prescribe running drills and interval sessions at specific intensities — for example, 100–120% of MAS. Training at or above MAS enhances speed endurance, sprint recovery, and fatigue resistance, which are crucial for football performance.

25. Cone drills are designed to improve agility, which is the ability to change direction quickly and efficiently while maintaining balance, coordination, and control. The effectiveness of cone drills is based on several key training principles:

- Specificity:

  Cone drills mimic sport-specific movements such as cutting, sprinting, and turning. This helps athletes transfer the agility developed in training directly to game situations.

- Progression:

  The drills should move from simple patterns (like straight-line sprints) to more complex ones (like zig-zag or T-drills). Gradual progression enhances coordination and minimizes injury risk.

- Technique:

  Proper body position, low center of gravity, and quick footwork are emphasized to improve movement efficiency and control during rapid direction changes.

- Intensity and Reaction

  Athletes are encouraged to perform drills at game-like intensity or respond to external cues (visual or verbal) to develop reactive agility and decision-making speed.

- Consistency and Repetition:

  Regular practice of cone drills builds muscle memory**,** speed, and movement accuracy, which are essential for improving agility performance.

26. Plyometric training is a form of exercise that involves rapid and repeated stretching and contracting of muscles to increase muscular power and explosiveness**.** It utilizes the stretch-shortening cycle (SSC) — where a muscle is first stretched (eccentric phase) and then quickly shortened (concentric phase) — to produce maximum force in minimal time.

    The fundamental purpose of plyometric training is to develop explosive strength, which enhances performance in activities requiring sudden bursts of power such as jumping, sprinting, or throwing. It improves the efficiency of neuromuscular coordination, allowing muscles to generate greater force quickly.

    Additionally, plyometric training strengthens tendons and ligaments, improves balance, agility, and speed, and enhances overall athletic performance.

27. When designing plyometric programs for youth, special care must be taken to ensure safety, proper technique, and gradual development of strength and coordination. Since children and adolescents are still growing, their bones, joints, and muscles are more vulnerable to excessive stress.

- Start with Low Intensity:
  Begin with basic, low-impact drills such **as** hops, skips, and jump-and-stick exercises to build coordination and confidence before progressing to advanced movements.
- Focus on Technique:
  Emphasize proper landing mechanics — soft knees, stable posture, and controlled movement — to reduce injury risk and encourage good motor habits.
- Progress Gradually:
  Increase the intensity, volume, and complexity slowly based on the child's ability, not age. Avoid maximal jumps or heavy loads too early.
- Use Safe Surfaces and Supervision:
  Perform exercises on soft, non-slip surfaces (grass, mats) and ensure qualified supervision to correct errors and monitor fatigue levels.
- Make It Fun and Functional:
  Incorporate games, relay races, or sport-like activities to keep the training enjoyable and engaging, promoting long-term adherence and development.

28. Plyometric training involves high-impact and explosive movements, which place significant stress on the muscles, joints, and tendons. Therefore, it is not suitable for certain populations where such stress could cause harm or aggravate existing conditions.

- **Individuals with Joint or Bone Injuries:**
  Those recovering from ankle, knee, hip, or spinal injuries should avoid plyometrics until fully rehabilitated, as impact forces can worsen damage.
- **People with Poor Strength Base or Fitness Level:**
  Beginners who lack adequate muscular strength, balance, or coordination are at higher risk of injury due to poor control during landing and jumping.
- **Elderly Individuals:**
  Due to reduced bone density, flexibility, and joint stability, older adults may experience strain or falls during explosive exercises.
- **Individuals with Obesity:**
  Excess body weight increases impact forces on joints, leading to a higher chance of ligament or tendon injuries.

- **People with Certain Medical Conditions:**
  Those with cardiovascular diseases, osteoporosis, arthritis, or post-surgical recovery should avoid plyometric training without medical clearance.